LESSON 7.7

Represent Polygons on the Coordinate Plane

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can draw polygons from their vertex coordinates on the coordinate plane, and use coordinates to find side lengths and solve perimeter and area problems.

WHY THIS MATTERS A polygon's vertices are given as coordinates: (-4.5, 3), (5, 3), (5, -3.75), and (-4.5, -3.75). What is its perimeter?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
vertex *Spanish: vértice*	A corner point of a polygon, named by an ordered pair on the coordinate plane.	named by an ordered pair
polygon *Spanish: polígono*	A closed figure made of line segments; on the plane you draw it by plotting the vertices and connecting them in order.	triangle, rectangle, square
perimeter *Spanish: perímetro*	The total distance around a polygon — the sum of all its side lengths.	add every side length
distance between vertices *Spanish: distancia entre vértices*	When two vertices share an x- or y-coordinate, the distance between them is the absolute difference of the other coordinates.	(2, 4) and (5, 4) share y, so the side is $\|5 - 2\| = 3$ units.
scale *Spanish: escala*	What one grid unit stands for in the real situation.	1 unit = 1 meter
Side length *Spanish: Longitud del lado*	The distance between two vertices that are joined by a side.	found with absolute value

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A rectangle has vertices (−2, 3), (5, 3), (5, −2) and (−2, −2). Each unit is 1 meter. Find its perimeter.

1. Top side: same y, so 2 + 5 = 7 m.
2. Right side: same x, so 3 + 2 = 5 m.
3. Add all four sides: 7 + 5 + 7 + 5 = 24.

Answer: Perimeter = 24 m

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

The shape is a rectangle, so the opposite sides match: two sides of 9.5 and two of 6.75.

1. _____
2. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- Vertices _____ and _____ share the _____ -coordinate, so that side runs _____ .
- The length of that side is _____ units because _____ .
- The perimeter is _____ units because I added _____ .
- This polygon is a _____ because _____ .

Word bank vertex · polygon · perimeter · side length · coordinate plane · absolute value · horizontal · vertical · rectangle · square · add · units

WATCH OUT!

A common mistake

Subtracting coordinates without the absolute value

LESSON 7.7

Represent Polygons on the Coordinate Plane

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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WHY THIS MATTERS A polygon's vertices are given as coordinates: (-4.5, 3), (5, 3), (5, -3.75), and (-4.5, -3.75). What is its perimeter?

PART 1 Apply it

real-world problems

1 **Circle the letter of the best answer. Show how you know.**

Vertices (-4.5, 3) and (5, 3) share y = 3. What is the distance between them?

(A) 0.5 units
(B) 1.5 units
(C) 8 units
(D) 9.5 units

HINT Which coordinate is the same? Which differs?

My work — one step per line	What I did to both sides

1. **Look** *What is done to the variable?*
2. **Undo** *Do the opposite — both sides.*
3. **Simplify** *One side at a time.*
4. **Check** *Put it back in.*

2 **Circle the letter of the best answer. Show how you know.**

Vertices (5, 3) and (5, -3.75) share x = 5. What is the distance between them?

(A) 0.75 units
(B) 5 units
(C) 6.75 units
(D) 8.75 units

HINT Which coordinate differs?

My work — one step per line	What I did to both sides

1. **Look** *What is done to the variable?*
2. **Undo** *Do the opposite — both sides.*
3. **Simplify** *One side at a time.*
4. **Check** *Put it back in.*

3 **Circle the letter of the best answer. Show how you know.**

What is the perimeter of the rectangle with side lengths 9.5 and 6.75 units?

(A) 16.25 units
(B) 26 units
(C) 32.5 units
(D) 64.125 units

HINT A rectangle has four sides.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

4 **Circle the letter of the best answer. Show how you know.**

Why are two of the terms multiplied by 2 in the perimeter calculation?

(A) Because perimeter always doubles the area

(B) Because there are two coordinates in each ordered pair

(C) It is a rounding rule

(D) A rectangle's opposite sides are equal, so each length appears on two sides

HINT Walk the rectangle: how many 9.5 sides?

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

To draw a polygon from a list of vertices, you:

(A) Plot only the first vertex

(B) Connect the points in any random order

(C) Add all the coordinates together

(D) Plot each ordered pair, then connect them in order with segments

HINT What did the deck say to do first?

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

6 **Answer in complete sentences.**

A square has vertices (1, 2), (6, 2), (6, -3), (1, -3). Find one side length and the perimeter.

SENTENCE STARTER The vertices ___ and ___ share ___ .
So one side is ___ , and the perimeter is ___ .

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Take a side you both put under 'Slanted.' Why does subtracting coordinates give the wrong length there, and what has to be true about two vertices for subtraction to work?

SENTENCE STARTER Subtracting works when the two vertices share ___, so for ___ I cannot ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 7.7

Represent Polygons on the Coordinate Plane

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
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Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can draw polygons from their vertex coordinates on the coordinate plane, and use coordinates to find side lengths and solve perimeter and area problems.

WHY THIS MATTERS A polygon's vertices are given as coordinates: (-4.5, 3), (5, 3), (5, -3.75), and (-4.5, -3.75). What is its perimeter?

PART 1 Apply it

real-world problems

1 **Circle the letter of the best answer. Show how you know.**

The park's marked vertices are 3 units apart and each unit is 100 feet. What is the park's width?

(A) 3 feet
(B) 30 feet
(C) 103 feet
(D) 300 feet

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

The park's area must be 240,000 sq ft and its width is 300 ft. What is its length?

(A) 720 feet
(B) 800 feet
(C) 80,000 feet
(D) 240,300 feet

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

3 **Circle the letter of the best answer. Show how you know.**

The length is 800 feet and each unit represents 100 feet. How many units long is the park on the map?

(A) 0.8 units
(B) 8 units
(C) 80 units
(D) 800 units

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

The known vertices of the park are at (2, 4) and (5, 4), and the length runs DOWNWARD 8 units. Where are the other two vertices?

(A) (2, -4) and (5, -4)
(B) (2, 12) and (5, 12)
(C) (2, -8) and (5, -8)
(D) (10, 4) and (13, 4)

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A rectangle has vertices (-3, 2), (4, 2), (4, -1), (-3, -1). What is its perimeter?

(A) 10 units
(B) 13 units
(C) 20 units
(D) 21 units

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

PART 2 Explain and correct

reasoning

6 **Answer in complete sentences.**

Sheng plans a zoo visit using a map where exhibits sit at grid points. Explain how identifying WHAT the problem asks (his shortest route? total distance?) changes what you compute.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Take a side you both put under 'Slanted.' Why does subtracting coordinates give the wrong length there, and what has to be true about two vertices for subtraction to work?

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LESSON 7.7

Represent Polygons on the Coordinate Plane

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
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Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can draw polygons from their vertex coordinates on the coordinate plane, and use coordinates to find side lengths and solve perimeter and area problems.

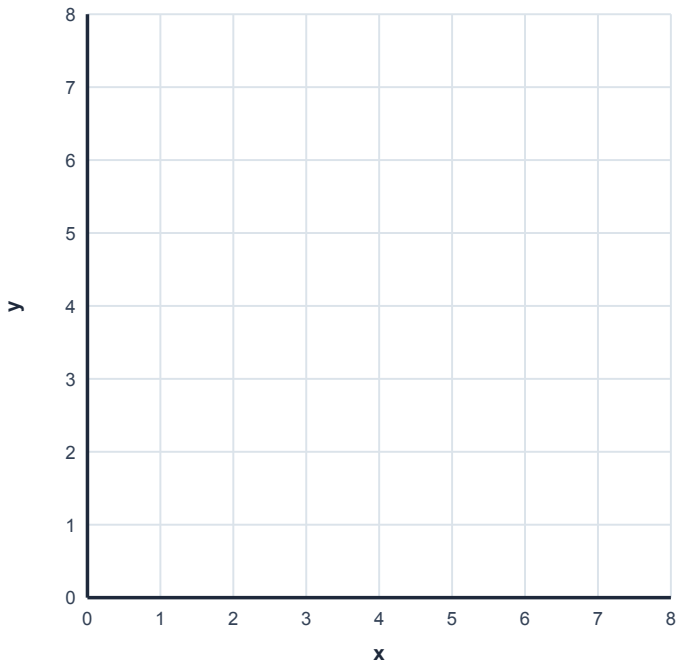
WHY THIS MATTERS A polygon's vertices are given as coordinates: (-4.5, 3), (5, 3), (5, -3.75), and (-4.5, -3.75). What is its perimeter?

PART 1 Understand the idea

models and meaning

1 **Plot and label each point on the grid.**

Plot the vertices of rectangle PQRS, then use the coordinates to find its perimeter.



SHOW YOUR THINKING

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Maria is choosing between two store spaces on a grid where each unit is 1 foot and each grid line is 5 feet apart. Space A is a rectangle 25 ft by 40 ft; Space B is 30 ft by 30 ft. Which has more floor area?

(A) Space A — 1,000 sq ft vs 900 sq ft

(B) Space B — 900 sq ft vs 850 sq ft

(C) They are equal

(D) Space B — because a square always beats a rectangle

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A square has vertices (a, b) and (a, b + s) on one side. Which operation gives the side length for ANY values?

(A) Add all four coordinates

(B) Subtract the y-coordinates and take the absolute value

(C) Multiply a × b

(D) Average the coordinates

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

4 **Answer in complete sentences.**

The park design used width from coordinates, then length from the area, then the scale to place the missing vertices. Design your own version: pick an area and a scale, mark two vertices, and show where the other two go.

WRITE YOUR ANSWER

5 **Answer in complete sentences.**

Check the deck's perimeter of 32.5 units a DIFFERENT way than adding the four sides in order.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Take a side you both put under 'Slanted.' Why does subtracting coordinates give the wrong length there, and what has to be true about two vertices for subtraction to work?

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